

Conditional *wh*-questions with VP Ellipsis

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Abstract

I discuss a construction I call *conditional wh-questions with VP Ellipsis*, which are root object *wh*-questions that appear with an *if*-clause, in which there is VP Ellipsis. Two interpretations of the elided object are allowed: a *wh*-bound variable or a narrow-scope indefinite. Structural diagnostics reveal that the interpretation of the elided object is correlated with the height of the *if*-clause: the *wh*-bound reading obtains iff the *if*-clause is attached inside the *wh*-question. The indefinite reading of the elided object is interesting because there the elided object apparently mismatches with its antecedent, a (reconstructed) *wh*-phrase, in two ways: they differ in their syntactic forms (a non-*wh* indefinite vs. a *wh*-phrase) as well as scope (a narrow-scope indefinite vs. a wide-scope existential quantification that scopes over the question operator). Combining a PF-deletion theory of ellipsis with a semantic identity condition and independently motivated mechanisms for interpreting *wh*-questions and indefinites, I develop an account of the height-interpretation correlation of the elided object that also explains why the elided object with an indefinite reading is allowed the mismatch that it is: through the appropriate analytical lens, the apparent mismatch is not genuine.

Keywords: Syntax, Semantics, VP Ellipsis, Conditionals

1 Introduction

This paper concerns a construction I call *conditional wh-questions with VP Ellipsis*, exemplified in (??) and (??) below.¹ Descriptively speaking, they are root object *wh*-questions that appear with an *if*-clause, which in turn contains an elided VP. For convenience, I will refer to them simply as *conditional wh-questions* whenever the presence or absence of VP Ellipsis is evident from context.

- (1) a. What museum would you visit if SALLY did Δ ?
- b. If you COULD Δ , what advice would you give to your younger self?

These examples are most naturally paraphrased as (??) and (??):

- (2) a. What museum x is such that you would visit x if Sally visited x ?
- b. If you could give advice to your younger self, what advice x is such that you would give x to your younger self?

These examples are distinguished by the interpretation of the elided object. Namely, (??) has a *bound reading*, where the elided object is interpreted as a variable bound by the *wh*-phrase, while (??) has an *indefinite reading*, where the elided object is interpreted as an indefinite.

Central to any theory of ellipsis is an identity condition, which must hold between an elided phrase and its antecedent for its ellipsis to be licensed. Examples of ellipsis where the elided phrase and its antecedent apparently mismatch in some respect are particularly informative, as they instruct us how to state the identity condition so that it tolerates the right differences. Many such examples have been observed in the literature; three illustrative examples are given in (??). In each example, the object in the antecedent VP VP_A and the object in the elided VP VP_E mismatch in some way.

- (3) a. *Name vs. pronoun* (?, ex. 63)
They [VP_A arrested Alex₃], though he₃ thought they wouldn't $\langle VP_E$ ~~arrest him₃~~ $\rangle$.
- b. *Negative polarity item vs. indefinite* (?, ex. 123,125)
John doesn't [VP_A have any horses], but Bill does $\langle VP_E$ ~~have some horses~~ $\rangle$.
- c. *Negative indefinite vs. indefinite* (?, ex. 103,104)
I could [VP_A find no solution], but Holly might $\langle VP_E$ ~~find a solution~~ $\rangle$.

Conditional *wh*-questions with indefinite readings (??) constitute a new data point of this kind; while the elided object is an indefinite, the antecedent object is a *wh*-phrase (or its trace). This is an example of a mismatch because indefinites and *wh*-phrases have very different syntactic properties. The primary goal of this paper is to

¹Notations in this paper: UPPERCASE denotes accented material. Δ denotes an ellipsis site. Elided material are ~~enclosed in angled brackets and struck through~~. A focus-marked constituent is indicated like _F[this]. Binding indices are subscripted on linguistic expressions that behave like variables (e.g. it_2 , t_3) and superscripted on those that behave like binders (e.g. $[\text{every student}]^4$, $[\text{which dog}]^5$). For types τ_1, τ_2 , the function type $\langle \tau_1, \tau_2 \rangle$ is abbreviated as $\tau_1\tau_2$. This abbreviation is right-associative: e.g., *est* is $\langle e, st \rangle$, not $\langle es, t \rangle$. In lambda-terms, variable binders will be subscripted with the type of the variable it binds (e.g. $\lambda x_e, \forall p_{st}, \exists f_{et}$). Metalinguistic constants are indicated with sans-serif font (e.g., *dog*, *believe*). For a type τ , D_τ is the set of semantic objects of type τ .

understand why VP Ellipsis allows this mismatch to take place. To do that, I suggest we first better understand the structures underlying the two readings of conditional *wh*-questions with VP Ellipsis.

This paper is structured as follows. In Section ??, I provide more examples of conditional *wh*-questions with VP Ellipsis. These examples clearly indicate a rough correlation between the linear position of the *if*-clause and the interpretation of the elided object: the bound reading prefers a final *if*-clause, while the indefinite reading prefers an initial *if*-clause. In Section ??, I use three diagnostics to show that the two readings result from different attachment heights of the *if*-clause during interpretation: the bound reading arises when the *if*-clause is interpreted as attached *low*, i.e., inside the *wh*-question, while the indefinite reading arises when the *if*-clause is interpreted as attached *high*, i.e., outside the *wh*-question. I illustrate how this fact helps us derive the different preferences for the linear position of the *if*-clause exhibited by the two readings.

In Section ??, I provide an analysis that explains why ellipsis is licensed in the two readings of conditional *wh*-questions only when their respective demands on their *if*-clauses' heights are met. Recall that the indefinite reading allows the ellipsis of a non-*wh* indefinite object, with a *wh*-phrase serving as its antecedent. I take the mismatch between the elided and antecedent objects in the indefinite reading to be a genuine syntactic mismatch, and resort to a theory of ellipsis licensing that requires the elided phrase and its antecedent to be semantically, not syntactically, identical. Specifically, I assume a semantic identity condition to hold between a pair of constituents known as *parallelism domains*, which must respectively contain the elided phrase and its antecedent. Any semantic difference between the parallelism domains can only be neutralized by focus-marking the constituent that is responsible for causing said difference; otherwise, ellipsis is unlicensed. Under this theory, an ellipsis analysis reduces to locating an appropriate pair of parallelism domains, in which any lexical mismatch is neutralized by focus-marking.

Ellipsis licensing in the bound reading is explained under this theory. The explanation applies generally to a configuration of VP Ellipsis known as *Co-binding*, which subsumes the bound reading. An account of the indefinite reading also seems within reach, especially because of the long observed semantic connection between *wh*-phrases and indefinites (?). However, the indefinite reading poses a challenge as it embodies a scopal mismatch: the antecedent object is a *wh*-phrase, which introduces wide-scope existential quantification indefinite scoping over the question operator denoted by the interrogative complementizer, but the elided indefinite object, as I will show, has a narrow-scope existential reading. In general, the ellipsis of an element that mismatches in scope with its antecedent is banned, and the semantic identity theory of ellipsis licensing I adopt also enforces this ban. I present a solution to this scopal mismatch by adopting a Skolemized choice function analysis of *wh*-phrases and indefinites. This helps us sidestep the scopal mismatch by modelling a narrow-scope indefinite as a wide-scope quantification over Skolemized choice functions.

My analysis predicts that we will find indefinite readings of an elided object anteceded by a *wh*-phrase even if the elided VP is hosted in clausal adjuncts other than conditionals. I discuss this prediction in Section ??. I conclude in Section ??.

The main contributions of this paper are to document a new case of mismatch tolerated by VP Ellipsis, namely the mismatch between an elided narrow-scope indefinite object and an antecedent *wh*-object, and to present an ellipsis analysis that derives the apparent connection between the height of the *if*-clause hosting the elided VP and the interpretation of the elided object.

2 Linear position correlates with reading

In this section, I provide more examples of conditional *wh*-questions with VP Ellipsis (Section ??) and point out a rough correlation between the linear position of the *if*-clause and the interpretation of the elided object: the bound reading prefers a final *if*-clause, while the indefinite reading prefers an initial *if*-clause (Section ??).

2.1 Examples of conditional *wh*-questions with VP Ellipsis

I provide a context for each example below in (??) and (??), as well as a paraphrase of its most accessible reading. The descriptions of these contexts are meant to be non-linguistic and thus are not good sources of antecedents for VP Ellipsis.

(4) *Examples with final if-clauses*

- a. (Context: you and I are grad students at a conference. You don't really like to talk to people, but I forced you to talk to many people anyways since I think that's important for networking purposes. You aren't too happy about what I did. I ask you sarcastically:
Who would you have willingly talked to if I hadn't forced you to?
Paraphrase: Which person x is such that you would have willingly talked to x if I hadn't forced you to talk to x ?)
- b. (Context: you and I both know John doesn't like to go to music festivals on his own, unlike all of his friends. He really likes to spend time with his friends, though. We also know there are music festivals that John is secretly interested in attending, and all it takes for John to actually go is for all of his friends to go to these festivals. I ask you, to find out what these festivals are:
Which music festival would John go to if all of his friends did?
Paraphrase: Which music festival x is such that John would go to x if all of his friends went to x ?)
- c. (Context: I know you like hanging out with Sally and you don't particularly like to go to museums. So I ask you:
What museum would you visit if Sally did?
Paraphrase: What museum x is such that you would visit x if Sally visited x ?)
- d. (Context: Mary is a grad student in linguistics who works on a lot of language families. Her advisor likes to tell grad students to work on as many language families as possible. In order to tease apart the language

families Mary originally had interest in from those she had to be convinced by her advisor to work on, I ask:)

What language family would Mary never have worked on if her advisor hadn't told her she should?

Paraphrase: What language family x is such that Mary would never have worked on x if her advisor hadn't told her she should work on x ?

(5) *Examples with initial if-clauses*

- a. (Context: you are a grad student, and I am your advisor. Our department asks advisors to give a letter grade to each grad student at the end of every semester for their performance this semester. I want your input to decide on your grade, so I ask you:)

If you had to, what grade would you give yourself given your performance this semester?

Paraphrase: If you had to give yourself a grade, what grade x is such that you would give yourself x given your performance this semester?

- b. (Context: you and I are reflecting on the major decisions we made in life. I ask:)

If you could, what advice would you give to your younger self?

Paraphrase: If you could give advice to your younger self, what advice x is such that you would give x to your younger self?

- c. (Context: Bill is a big fan of Kraftwerk. You and I are joking about his obsession with Kraftwerk. I ask you:)

If Bill were asked to, which Kraftwerk song do you think he would play?

Paraphrase: If Bill were asked to play a Kraftwerk song, which Kraftwerk song x is such that you think he would play x ?

- d. (Context: you and I are grad students in the same PhD program. Having never read a single word of the grad student handbook, we are told by our advisors to check the handbook to see if we have completed our course requirements. In case we haven't, we wonder what is the easiest course we can take to clear the requirement. I ask you:)

If the handbook says we have to, which class should we take next semester?

Paraphrase: If the handbook says we have to take some class next semester, which class x is such that we should take x next semester?

The first group of examples, (??), all feature final *if*-clauses. They are all paraphrased as having a bound reading of the elided object. The second group of examples, (??), all feature initial *if*-clauses. They are all paraphrased as having an indefinite reading.

2.2 Linear position correlates with reading

The bound readings of the examples with final *if*-clauses (??) become severely degraded when their *if*-clauses are fronted to an utterance-initial position, as shown in (??):

(6) *Bound reading examples (??) with initial if-clauses instead*

- a. If I hadn't forced you to, who would you have willingly talked to? (*bound)
- b. If all of John's friends did, which music festival would he go to? (*?bound)
- c. If Sally did, what museum would you visit? (*bound)
- d. If Mary's advisor hadn't told her she should, what language family would she never have worked on? (??bound)

The indefinite readings of the examples with initial *if*-clauses (??) also degrade when their *if*-clauses are brought to an utterance-final position, as shown in (??):

- (7) *Indefinite reading examples (??) with final if-clauses instead*
 - a. What grade would you give yourself given your performance this semester, if you had to? (??indefinite)
 - b. What advice would you give to your younger self, if you could? (?indefinite)
 - c. Which Kraftwerk song do you think Bill would play, if he were asked to? (*?indefinite)
 - d. Which class should we take next semester, if the handbook says we have to? (*?indefinite)

Thus, there is a clear correlation between the linear position of the *if*-clause and the interpretation of the elided object:

- (8) *Correlation between linear position and reading*
 - A bound reading is better with a final *if*-clause than with an initial *if*-clause.
 - An indefinite reading is better with an initial *if*-clause than with a final *if*-clause.

Note that this correlation is rough and leaky, and it is intentionally stated as such. A bound reading isn't categorically impossible with an initial *if*-clause (e.g., (??)), nor is an indefinite reading categorically impossible with a final *if*-clause (e.g., (??), (??)); in fact, some speakers find the indefinite reading quite accessible with final *if*-clauses (??). Even those who don't accept the examples in (??) tend to find them better than the bound reading examples with initial *if*-clauses (??). This correlation, as well as the acceptability difference between (??) and (??), will receive explanations in Section ??.

The correlation (??) only states that the two readings prefer different linear positions of the *if*-clause; it shouldn't lead the reader to expect one of the two readings to consistently arise in just any conditional *wh*-question with VP Ellipsis. The availability of the two readings are controlled by other independent factors, possibly most importantly by pragmatic plausibility.

3 Height determines reading

The goal of this section is to find an explanation for the correlation (??) in Section ??, which linked the linear position of the *if*-clause to the interpretation of the elided object in conditional *wh*-questions. A relevant insight from the literature on the syntax of conditionals is that the many behaviors of *if*-clauses can be explained by thinking

about the heights at which they are attached to their matrix clauses. With this in mind, I propose that the two readings of conditional *wh*-questions with VP Ellipsis require their *if*-clauses to be attached at different heights in their logical forms (LFs) (??):

(9) *Claim: height determines reading*

An LF where the *if*-clause is interpreted *low*, i.e., inside the *wh*-question, can be a LF for the bound reading but not for the indefinite reading.

An LF where the *if*-clause is interpreted *high*, i.e., outside the *wh*-question, can be a LF for the indefinite reading but not for the bound reading.

In other words, a conditional *wh*-question with VP Ellipsis can have the bound reading only if the *if*-clause is contained within the *wh*-question, and it can have the indefinite reading only if the *if*-clause isn't contained in the *wh*-question. The configurations for the bound (??) and indefinite (??) reading examples at the beginning of this paper are illustrated in (??) and (??):

- (10) a. [_{wh-question} [What museum]¹ would you visit _{*t*₁} [_{if-clause} if SALLY did ~~visit it_T~~]]?
b. [_{if-clause} If you COULD ~~give advice to your younger self~~], [_{wh-question} what advice would you give to your younger self]?

The first clause of (??) isn't surprising, because the bound reading, as its name suggests, requires the *wh*-phrase to bind the elided object, and this binding isn't possible unless the *if*-clause is contained in the *wh*-question. The second clause, which concerns the indefinite reading, is less trivial and needs to be argued for. Using diagnostics proposed in the conditional literature, I offer three arguments in order to defend the claim (??) (Section ??). Then, I show how the correlation (??) follows from (??) (Section ??). I sketch tentative LFs for the two readings of conditional *wh*-questions, i.e., more refined versions of (??) and (??) with enough details for us to think about ellipsis licensing (Section ??). Finally, I defend a tacit assumption I have been making so far, namely the assumption that the bound and indefinite readings require distinct LFs (Section ??).

3.1 Arguments for (??)

3.1.1 Condition C

A fundamental result from the study on the syntax of conditionals is that utterance-initial and utterance-final *if*-clauses are attached at different heights relative to their matrix clauses. This is often argued for using Condition C effects. For example, ?? show that when the *if*-clause is initial, its subject can corefer with the matrix subject no matter if the *if*-clause subject (??) or the matrix subject (??) is a name. Since there are no Condition C violations, they conclude that an initial *if*-clause is adjoined to the matrix clause (??).

- (11) a. If Mary₁ is hungry, she₁ yells at Bill. (?, ex. 21b)
b. If she₁ is hungry, Mary₁ yells at Bill. (?, ex. 21c)

- (12) [_{if-clause} If Mary is hungry], [_{matrix} she yells at Bill].

On the other hand, if the subject of a final *if*-clause is a name, then it can't corefer with the matrix subject (??), but it can corefer with the matrix object (??). They conclude that a final *if*-clause is adjoined somewhere between the TP and the VP of the matrix clause (??).

- (13) a. *She₁ yells at Bill if Mary₁ is hungry. (?, ex. 21a)
 b. Bill visits her₁ if Mary₁ is sick. (?, ex. 22a)
- (14) [_{matrix} ... [TP She [_{VP} yells at Bill] [_{if-clause} if Mary is hungry]]].

What do Condition C tests tell us about the structures of conditional *wh*-questions? Consider the indefinite reading examples (??), (??). The lack of a Condition C violation with an initial *if*-clause (??) is expected, as we know initial *if*-clauses are attached high (??). Surprisingly, there is no Condition C violation even with a final *if*-clause (??), which leads us to conclude that the *if*-clause is question-external in this example as well (??). Thus, we know that an indefinite reading LF allows its *if*-clause to be interpreted as attached high, no matter its linear position.

- (15) a. If Sally₂ COULD (~~give advice to her₂ younger self~~), [what advice]¹ would she₂ give *t*₁ to her₂ younger self?
 b. [_{if-clause} If Sally₂ COULD (~~give advice to her₂ younger self~~)], [_{wh-question} [what advice]¹ would she₂ give *t*₁ to her₂ younger self]?
- (16) a. [What advice]¹ would she₂ give *t*₁ to her₂ younger self, if Sally₂ COULD (~~give advice to her₂ younger self~~)?
 b. [_{wh-question} [What advice]¹ would she₂ give *t*₁ to her₂ younger self], [_{if-clause} if Sally₂ COULD (~~give advice to her₂ younger self~~)]?

Why does (??) but not (??) trip Condition C, even though both examples have final *if*-clauses? I suggest that there is a general tendency to parse a final *if*-clause as attached low (?), which leads to the ungrammaticality of (??). This parsing preference can be countered by other factors, such as the pressure from the indefinite reading of conditional *wh*-questions to parse the *if*-clause as attached high, or prosodic clues indicating that the *if*-clause is separated from the matrix clause. Both of these factors are in effect when parsing (??); the comma before the *if*-clause signals the prosodic separation. Thus, (??) is grammatical. Prosodic clues alone can be enough to avoid a Condition C-violating parse, as shown in (??):

- (17) ?[Which restaurant]¹ would she₂ usually go to *t*₁, if Sally₂ wanted some good Vietnamese food?

Turning to the bound reading, we see that (??) is ungrammatical. This has to be a Condition C violation, because the grammatical (??) forms a minimal pair with (??). Thus, we know a bound reading LF doesn't allow its *if*-clause to be interpreted high (??).

- (18) a. *[What book]¹ would he₂ read *t*₁ if John₂ was told to (~~read it_T~~)?
 b. [What book]¹ would John₂ read *t*₁ if he₂ was told to (~~read it_T~~)?

- c. [$_{wh\text{-question}}$ [What book]¹ would John₂ read t_1 [$_{if\text{-clause}}$ if he₂ was told to ~~read it~~]]?

3.1.2 Quantifier binding

Another test that diagnoses c-command relations like Condition C tests is binding. As (??) below shows, a quantificational subject of the matrix clause can bind into a final *if*-clause. This is expected, since we know final *if*-clauses can be contained within their matrix clauses. Surprisingly, this kind of binding can target an initial *if*-clause as well (??).

- (19) a. [No woman]² would be proud of her₂ paper if she₂ only had three days to work on it.
b. If she₂ only had three days to work on it, [no woman]² would be proud of her₂ paper.

The grammaticality of (??) can be explained if we assume that its initial *if*-clause is derived by fronting a low *if*-clause; in other words, the underlying structure of (??) is (??). This fronting is reconstructed at LF, allowing the binding in question to proceed as it is in (??).

When there is independent pressure that forces *if*-clauses to be interpreted as attached high, this kind of binding is predicted to be impossible. Indeed, this is exactly what we see for the conditional *wh*-question examples (??) and (??) with indefinite readings: the matrix subject can't bind into the *if*-clause, no matter its linear position. This again confirms that the indefinite reading requires a high *if*-clause.

- (20) a. *[What advice]¹ would [no woman]² give t_1 to her₂ younger self, if she₂ could ~~give advice to her₂ younger self~~?
b. *If she₂ could ~~give advice to her₂ younger self~~, [what advice]¹ would [no woman]² give t_1 to her₂ younger self?

On the other hand, the bound reading allows the matrix subject to bind into the *if*-clause (??), just as the *wh*-phrase is able to bind into the *if*-clause. The *if*-clause of a bound reading is low.

- (21) [What book]¹ would [no man]² read t_1 if his₂ parents told him₂ not to ~~read it~~?

3.1.3 If-then

Another environment that forces a high-attachment interpretation of *if*-clauses is *if-then* conditionals, i.e., sentences of the form *if P then Q*. The *if*-clause in this construction is commonly analyzed as base-generated as an adjunct to the *then Q* phrase (????). For example, ? posit the LF (??) for the example (??).

- (22) a. If it rains, then I think we should stay at home.
b. [If it rains]², [then₂ λ3 I think [we should stay at home t_3]].

Example (??) also confirms that the *if*-clause in this construction is interpreted high; the matrix subject can't bind into the *if*-clause (compare with (??), where this kind of binding is possible).

- (23) *If she₂ only had three days to work on it, then [no woman]² would be proud of her paper.

We can combine *if-then* conditionals and conditional *wh*-questions: these are conditional *wh*-questions with an initial *if*-clause and a *then* prefixed to the *wh*-question part. Since these are instances of *if-then* conditionals, their *if*-clauses must be interpreted high. If we are right that an indefinite reading requires a high *if*-clause and a bound reading requires a low *if*-clause, then we would expect the indefinite reading but not the bound reading to be possible with these *if-then* flavors of conditional *wh*-questions. This is exactly borne out in (??):

- (24) a. If you could ~~give advice to your younger self~~, then [what advice]¹ would you give *t*₁ to your younger self?
 b. *If Sally did ~~visit it₁~~, then [what museum]¹ would you visit *t*₁?

Thus, evidence from Condition C tests, binding and *if-then* conditionals tells us that the bound reading of conditional *wh*-questions requires a low *if*-clause at interpretation, and the indefinite reading requires a high *if*-clause at interpretation. This is exactly the claim stated in (??).

3.2 How (??) follows from (??)

Having understood that the bound and indefinite readings demand their *if*-clauses be interpreted as attached at different heights, we are now ready to derive the correlation (??) from Section ??, i.e., to explain why the two readings prefer different linear positions of the *if*-clause.

First, consider a conditional *wh*-question with a bound reading. By (??), we know its *if*-clause is interpreted at a low, question-internal position. Such an *if*-clause is canonically realized finally and prosodically integrated into the *wh*-question that contains it. This is why the bound reading prefers a final *if*-clause. If this *if*-clause is fronted to an initial, question-external position and linearized initially, it must be reconstructed to its base position prior to interpretation. This reconstruction is the source of the difficulty of the bound reading with an initial *if*-clause.

Next, consider a conditional *wh*-question with an indefinite reading. By (??), we know its *if*-clause is interpreted at a high, question-external position. Such an *if*-clause is canonically linearized either initially or finally, but in either case, the *if*-clause is realized as prosodically separated from the *wh*-question it is adjoined to. An initial *if*-clause is unambiguously parsed as a high *if*-clause; this is why the indefinite reading prefers an initial *if*-clause. A final *if*-clause is ambiguous between a low parse and a high parse. While a low parse is generally preferred, interpretive pressures as well as prosodic clues can guide the comprehender towards a high parse, as discussed in Section ??. Depending on how costly it is for a given comprehender to reach the high parse, the unacceptability of the indefinite reading with a final *if*-clause can vary. For some, it is completely acceptable; for others, it is markedly worse compared to

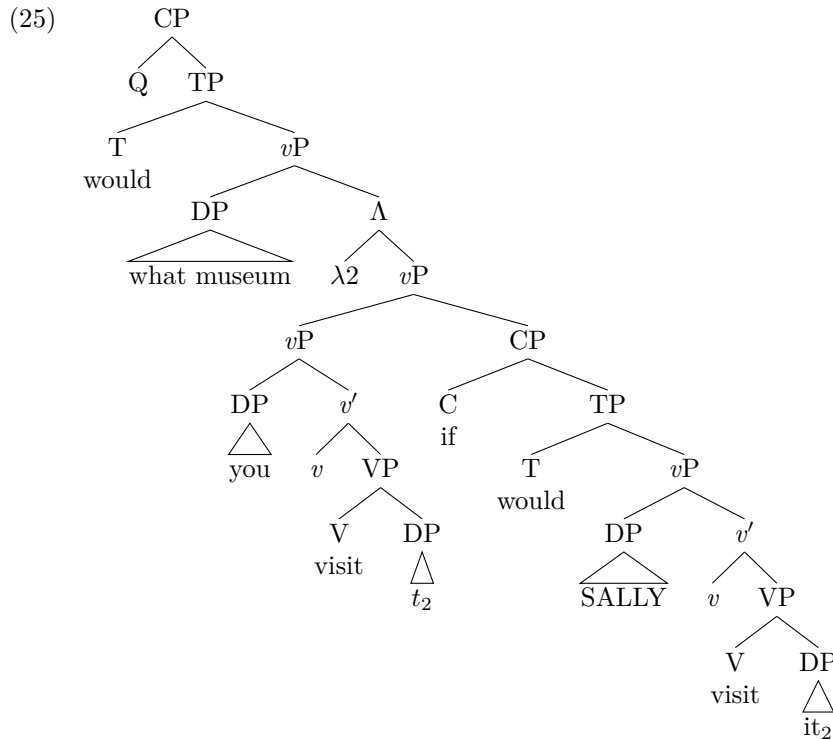
the indefinite reading with an initial *if*-clause. However, an indefinite reading with a final *if*-clause is always better than a bound reading with an initial *if*-clause. This is because an indefinite reading requires a high *if*-clause, which is a possible parse of a final *if*-clause, but a bound reading requires a low *if*-clause, which isn't a possible parse of an initial *if*-clause; it must be reconstructed.

This concludes the explanation for the correlation (??), as well as the acceptability differences between (??) and (??).

3.3 Tentative LFs for the bound and indefinite readings

I now propose LFs for the bound and indefinite readings of conditional *wh*-questions with VP Ellipsis that respect (??), our discovery that their *if*-clauses are interpreted at different heights. For concreteness, I will focus on the bound (??) and indefinite reading examples (??) that are hopefully familiar by now.

The LF for the bound reading example (??) is given in (??). For simplicity, I assume all head and A-movement to be syntactically reconstructed at LF.



The LF (??) sneaks in two assumptions that need to be defended. The first one concerns the nature of the elided object, which is assumed to be a pronoun semantically bound by *wh*-movement. Since this object is elided, how do we know it's a pronoun and not something else, e.g., a parasitic gap? ?, who studies a group of relevant constructions, shows that the pronoun analysis prevails over the parasitic gap analysis in two ways. First, conditional *wh*-questions with the bound reading allows the *wh*-phrase to

bind an elided object inside an island (??). This binding should only be possible if the elided object is a pronoun (??) and not a parasitic gap (??), since parasitic gaps show island effects.

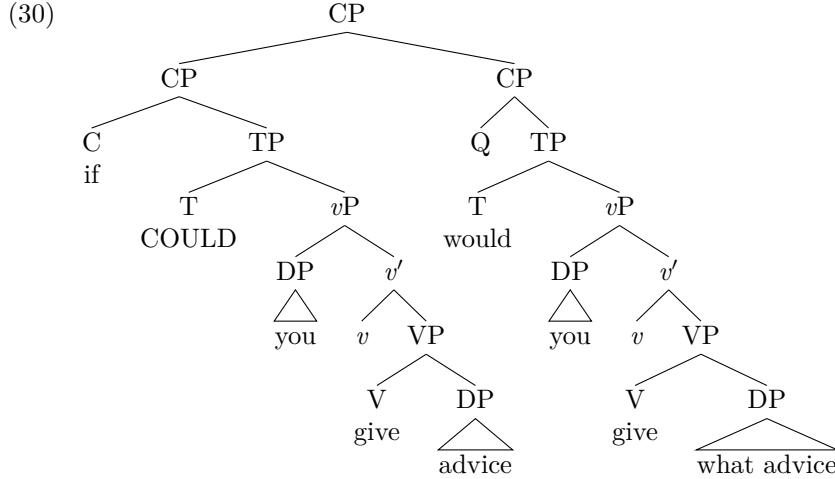
- (26) [Which article]¹ did you file t_1 without asking who else had Δ ? (?, ex. 25b)
- (27) a. [Which article]¹ did you file t_1 without asking who else had $\langle \text{read-it}_T \rangle$?
b. *[Which article]¹ did you file t_1 without asking who else had $\langle \text{read-pg}_T \rangle$?
(?, ex. 25a)

Second, the elided object in the bound reading doesn't show strong crossover effects (??). In this sense, the elided object patterns with a pronoun (??) and not with a parasitic gap (??).

- (28) [Which candidate]¹ did Maureen support t_1 because he₁ asked her to Δ ? (?, ex. 23)
- (29) a. [Which candidate]¹ did Maureen support t_1 because he₁ asked her to $\langle \text{support-him}_T \rangle$? (?, ex. 26)
b. *[Which candidate]¹ did Maureen support t_1 because he₁ asked her to $\langle \text{support-pg}_T \rangle$? (?, ex. 29)

The second assumption in the LF (??) concerns the *wh*-phrase *what museum*: it moves successive-cyclically through [Spec; vP] to [Spec; CP], and it is reconstructed to its intermediate position [Spec; vP]. There is independent evidence that objects move to [Spec; vP] (??), as well as evidence specific to *wh*-objects. The *wh*-phrase has to be reconstructed to [Spec; vP] because reconstruction to its base position would prevent it from binding the elided object, while not reconstructing at all would result in the elided object being bound from [Spec; CP], an A'-position. Allowing pronominal binding from A'-positions costs us popular accounts of crossover effects, specifically those whereby pronominal binding is only possible from A-positions (??), an undesirable outcome.

The LF for the indefinite reading example (??) is given in (??). Since the *wh*-phrase doesn't bind the elided object in (??), we are not bound by the earlier concerns that forced me to make specific decisions on the successive-cyclicity and the reconstruction of *wh*-movement for the bound reading LF (??). I thus remain agnostic as to these two aspects of *wh*-movement in the indefinite reading LF at this point. I represent the *wh*-phrase as reconstructed to its base position in (??) in order to prepare the reader for the ellipsis analysis in Section ??, which requires the *wh*-phrase to be interpreted below the interrogative complementizer Q.



Note that (??) and (??) are tentative LFs for the bound and indefinite readings designed with all the information about their structures we have obtained so far. They will be refined in Section ?? where I attempt to provide an ellipsis analysis.

3.4 Why do we need distinct LFs for the two readings?

So far, I have assumed that the bound and indefinite readings of conditional *wh*-questions are generated by distinct LFs. One might imagine an alternative analysis that only generates a LF for one of the two readings, from which the other reading follows by means of entailment, inference, etc.. Before I proceed to discuss my ellipsis analysis, I will rule out this alleged alternative explanation.

There are scenarios that make potential answers to the two readings of the same conditional *wh*-question different, and both readings are attested in such scenarios. An example is given in (??). The two conditional *wh*-questions in (??) and (??) are a minimal pair that only differs by the interpretation of the elided object, and they give rise to different felicitous responses. As a matter of fact, the two conversations (??) and (??) are both judged felicitous. This shows that both readings of the same conditional *wh*-question are possible given the same context (??), a difficult fact to account for without generating distinct LFs for the two readings.

- (31) Context: Bill is a big fan of Kraftwerk. He also doesn't like being told what to do. Any time someone asks him to play a specific Kraftwerk song, he always plays a different song from what was requested. But if you allow him to play one song of his choice, he always plays his favorite song, "Autobahn."
- (32) a. Person 1: [Which Kraftwerk song]¹ would Bill₂ play *t*₁ if he₂ were asked to ~~play it~~?
 Person 2: No song.
- b. Person 1: If he₂ were asked to ~~play a Kraftwerk song~~, [which Kraftwerk song]¹ would Bill₂ play *t*₁?
 Person 2: "Autobahn."

4 Analysis

The goal of this section is to produce an analysis of VP Ellipsis that derives the claim (??) from Section ???. That is, we want to know why ellipsis is licensed in each of the two readings of conditional *wh*-questions iff their *if*-clauses are interpreted at the correct heights.

In this paper, I will work with the PF-deletion framework of ellipsis licensing theories (?????????), whereby the phonological deletion of an elided constituent *E* is licensed by its satisfaction of an identity condition with some antecedent constituent *A*. A major debate in the literature concerns the nature of this identity condition: does it require *E* and *A* to be identical in their syntax, or in their semantics, or both?

Conditional *wh*-questions are particularly relevant in this debate. The familiar examples, (??) and (??), are repeated below. The *wh*-phrase in (??) is reconstructed.

(??) *Bound reading*

[What museum]² would you [_{VP_A} visit *t*₂] if SALLY did $\langle \text{VP}_{\overline{E}} \text{visit it}_2 \rangle$?

(??) *Indefinite reading*

If you COULD $\langle \text{VP}_{\overline{E}} \text{give advice to your younger self} \rangle$, would you [_{VP_A} give what advice to your younger self]?

In (??), the elided object is a pronoun, but its antecedent is a *wh*-trace. In (??), the elided object is an indefinite, which is presumably not *wh*, while its antecedent is a *wh*-phrase (or a *wh*-trace; the point still stands). If the elided object were *wh*, we would expect an interrogative licenser somewhere in the *if*-clause, but there is none. In both examples, the elided and antecedent objects take syntactically different forms (traces vs. pronouns, *wh*-phrases vs. indefinites). Yet, VP Ellipsis apparently doesn't care about this difference.

As I stated in Section ??, the indefinite reading adds to the growing list of apparent syntactic mismatches between certain kinds of DPs tolerated by VP Ellipsis, some of which are known as *vehicle change* effects (?). These examples all challenge a strict syntactic identity condition. Three examples are illustrated in (??), repeated below.

(??) a. *Name vs. pronoun* (?, ex. 63)

They [_{VP_A} arrested Alex₃], though he₃ thought they wouldn't $\langle \text{VP}_{\overline{E}} \text{arrest him}_3 \rangle$.

b. *Negative polarity item vs. indefinite* (?, ex. 123,125)

John doesn't [_{VP_A} have any horses], but Bill does $\langle \text{VP}_{\overline{E}} \text{have some horses} \rangle$.

c. *Negative indefinite vs. indefinite* (?, ex. 103,104)

I could [_{VP_A} find no solution], but Holly might $\langle \text{VP}_{\overline{E}} \text{find a solution} \rangle$.

It is possible to maintain a syntactic identity condition even when faced with such cases of syntactic mismatch. For example, one can weaken the identity condition so that it would be satisfied by partial identity between the syntactic structures of the mismatching DPs. This is exactly ?'s analysis of examples of the (??) type. Assuming that reflexives and definite DPs like names structurally contain pronouns, he proposes a syntactic identity condition, which he calls *structural recoverability*, that allows

reflexives and names to antecede an elided pronoun in VP Ellipsis via partial syntactic identity, accounting for (??).

A completely different reaction is to adopt a semantic identity condition. This looks promising, because the mismatched DPs in (??) can reasonably be construed as having identical denotations: a name and a coreferential pronoun have the same denotation (??), an NPI and a plain indefinite could be said to make equal semantic contributions (??), and a negative indefinite is often considered to be just a plain indefinite accompanied by sentential negation (??). This optimism might apply to the indefinite reading of conditional *wh*-questions as well: a *wh*-phrase and an indefinite are often thought to both denote existential quantification. This treatment is motivated by the cross-linguistic generalization that *wh*-phrases and indefinites share morphological makeup ?.

In this section, I provide an ellipsis analysis for conditional *wh*-questions based on ?'s theory of ellipsis licensing that uses a semantic identity condition. I first describe her theory (Section ??), and apply it to the bound reading of conditional *wh*-questions (Section ??). On the other hand, the indefinite reading of conditional *wh*-questions doesn't receive an adequate explanation. I provide an adequate analysis by taking indefinites and *wh*-phrases to denote Skolemized choice functions (Section ??).

4.1 ? without a syntactic identity condition

In this paper, I work with an adapted version of ?'s theory of ellipsis licensing, which originally consists of three parts: a syntactic identity condition on the elided VP and its antecedent, a semantic identity condition (??) on a pair of constituents that respectively contain the elided and antecedent VPs, and a constraint (??) that blocks "meaningless coindexing," a notion to be clarified later. For reasons elaborated earlier, the syntactic identity condition incorrectly rules out any case of syntactic mismatch tolerated by VP Ellipsis, including, most importantly, the indefinite reading of conditional *wh*-questions. There is also other evidence that the syntactic identity condition is too strong (?). Thus, I will drop it. I will postpone the discussion of (??) to Section ??, where its relevance will be obvious.

(33) *Focus condition (semantic identity condition)*

An elided VP must be contained in a constituent *E* which contrasts with a constituent *A*.

(34) *No Meaningless Coindexing*

If an LF contains an occurrence of a variable *v* that is bound by a node α , then all occurrences of *v* in this LF must be bound by the same node α .

That leaves us with just the semantic identity condition (??) to discuss. This is also known as the *focus condition*, and the constituents *E* and *A* are often referred to as *parallelism domains*. The focus condition (??) depends on a definition of contrast; an intuition for an appropriate definition is given in (??).

(35) *E contrasts with A* iff *E* and *A* have equal denotations up to any focus-marked parts in *E*.

According to (??), if E and A don't have equal denotations, then they don't contrast unless all the parts of E responsible for the semantic inequality are focus-marked. Consider (??), an example by ?. At LF, the indefinite subjects must be reconstructed to VP-internal positions on a narrow-scope reading (??). Consequently, the elided and antecedent VPs will have different denotations, and they will not qualify as parallelism domains on a definition of contrast that requires strict semantic identity. However, observe that the semantic difference between the two VPs ultimately stems from the choice of different lexical nouns for the reconstructed subjects in the VPs. The subject of the elided VP, *a DRAGON*, is accented, which we may assume to be the result of focus-marking. Such focus-marking, according to (??), neutralizes the semantic difference between the two VPs. Thus, the two VPs contrast, and ellipsis is correctly predicted to be licensed.

- (36) a. A unicorn was hoped for, and a DRAGON was $\langle \text{hoped for} \rangle$, too. (?, *ex. 98a*)
 b. Was $\boxed{A \text{ hoped for } [a \text{ unicorn}]}$, and was $\langle \text{hoped for } \mathbf{F}[a \text{ DRAGON}] \rangle$, too.

This ellipsis account also exemplifies a common strategy for explaining the presence of accented material in examples of VP Ellipsis: to attribute it to an instance of focus-marking that is critical for licensing ellipsis. Generally speaking, an instance of focus-marking is considered to be allowed only when it is necessary for satisfying other inviolable grammatical constraints, e.g. ellipsis licensing (?). In the case of (??), without the focus-marking on *a DRAGON*, a suitable pair of parallelism domains cannot be found in LF (??), and thus the ellipsis would not be licensed. I will employ this strategy in my own analyses of conditional *wh*-questions in Sections ?? and ??.

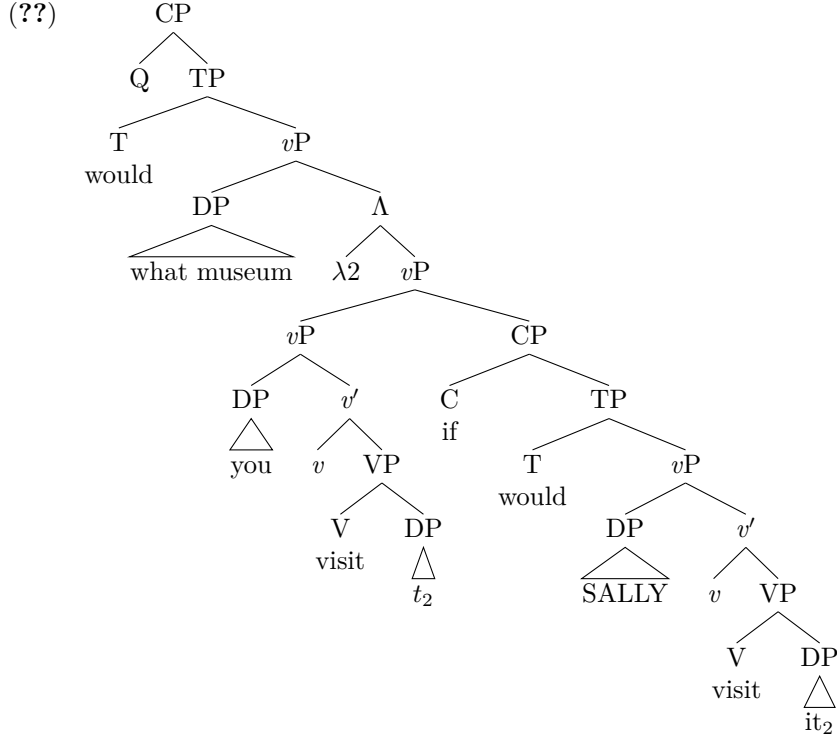
The intuition (??) has received numerous implementations in the literature, to various degrees of formality and complexity. Here, I think it suffices to adopt ?'s formulation (??):

- (37) E contrasts with A iff the ordinary semantic value $\llbracket A \rrbracket^g$ is a member of the focus semantic value $\llbracket E \rrbracket_f^g$ for all assignments g .
 (38) The focus semantic value $\llbracket [X \dots \mathbf{F}[Y] \dots] \rrbracket_f^g$ is the set consisting of all ordinary semantic values $\llbracket [X' \dots Y' \dots] \rrbracket^g$, where $\llbracket Y' \rrbracket^g$ is a semantic object with the same type as $\llbracket Y \rrbracket^g$ and X' is identical to X except that Y' replaces $\mathbf{F}[Y]$.

4.2 The bound reading

The theory of ellipsis licensing outlined in Section ?? successfully accounts for the bound reading of conditional *wh*-question. Recall that the goal is to explain the first part of the claim (??) from Section ??, i.e., why the bound reading is possible iff the *if*-clause is interpreted at a low, question-internal position. We won't commit to any specific theory of question semantics while we discuss the bound reading.

First, I will show that ellipsis is predicted to be licensed in a bound reading LF with a low *if*-clause (??), repeated below.



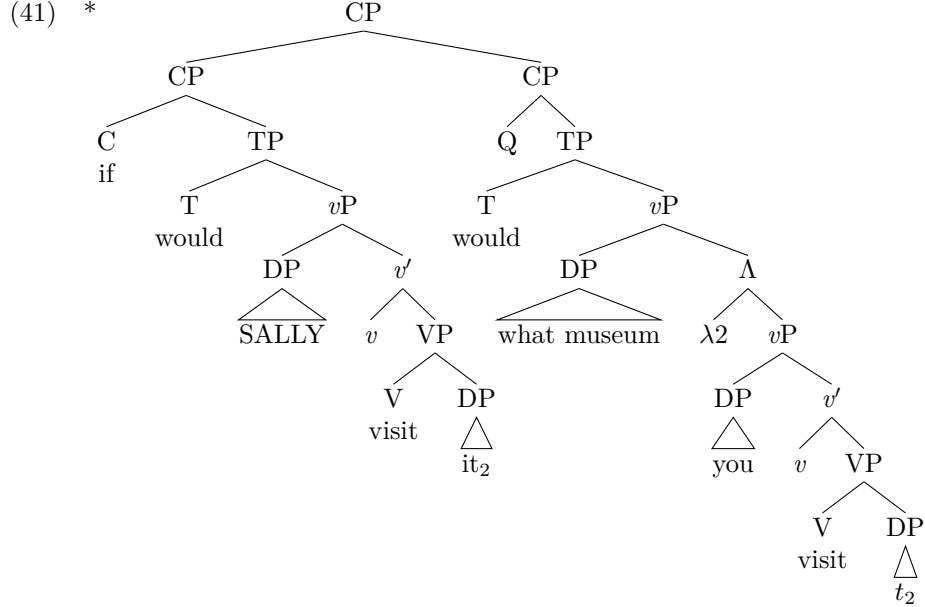
Notice that the nature of the bound reading forces the elided and antecedent objects to be coindexed. This means the elided and antecedent VPs have equal denotations under all assignments (??). Thus, the elided VP and the antecedent VP may serve as parallelism domains that satisfy the focus condition, and ellipsis would be correctly predicted to be licensed.

$$\begin{aligned}
 (39) \quad \llbracket A \rrbracket^g &= \llbracket \text{visit } t_2 \rrbracket^g = \text{visit}(g(2)) \\
 \llbracket E \rrbracket_f^g &= \llbracket \langle \text{visit } t_2 \rangle \rrbracket_f^g = \{\text{visit}(g(2))\} \\
 &\text{Thus, } \llbracket A \rrbracket^g \in \llbracket E \rrbracket_f^g \text{ for all } g.
 \end{aligned}$$

However, this leaves unexplained why there is accent on *SALLY*. We can apply the strategy I described in Section ??: argue that this accent results from focus-marking on *SALLY* that is necessary for licensing ellipsis. Then, we need to prevent constituents that don't include *SALLY* in (??) from being the parallelism domain E . This is possible if we assume that parallelism domains have to be as large as v Ps. Then, the v Ps *SALLY* $\langle \text{visit } t_2 \rangle$ and *I visit* t_2 qualify as parallelism domains, but the difference between their subjects has to be neutralized by focus-marking. The satisfaction of the focus condition is illustrated in (??), where D_e is the set of all semantic objects of type e . Focus-marking on the subject is then predicted to be obligatory.

$$\begin{aligned}
 (40) \quad \llbracket A \rrbracket^g &= \llbracket \text{I visit } t_2 \rrbracket^g = \text{visit}(g(2))(\text{sally}) \\
 \llbracket E \rrbracket_f^g &= \llbracket \text{F}[SALLY] \langle \text{visit } t_2 \rangle \rrbracket_f^g = \{\text{visit}(g(2))(x) \mid x \in D_e\} \\
 &\text{Thus, } \llbracket A \rrbracket^g \in \llbracket E \rrbracket_f^g \text{ for all } g.
 \end{aligned}$$

Next, let us turn to a bound reading LF with a high *if*-clause. As I stated in Section ??, such a LF *a priori* does not exist, if by a “bound reading LF” we mean one where the *wh*-phrase binds the elided object. But what if we are instead referring to a LF where the elided and antecedent objects are coindexed? A bound-reading-in-that-sense LF with a high *if*-clause certainly exists, and it is given in (??). It turns out that exactly the same argument we used for (??) can be applied to predict ellipsis licensing for (??). The same VPs (or *v*Ps) are parallelism domains. Such a LF would give rise to a non-*wh*-bound, referential reading of the elided object, paraphrased in (??), an ungrammatical case of ellipsis. Thus, this prediction is wrong.



- (42) *If SALLY would (visit it(= a contextually salient place)), what museum would you visit?

What we are facing is a specific instance of a well-known problem: an identity condition for ellipsis that depends on variable indices will overgenerate ellipsis when variables are coindexed coincidentally (?). Heim’s solution is to posit a constraint on LF that bans such meaningless coindexations (??), repeated below.

(??) *No Meaningless Coindexing*

If an LF contains an occurrence of a variable *v* that is bound by a node α , then all occurrences of *v* in this LF must be bound by the same node α .

Variables (i.e., traces, pronouns) are identified by their indices. The undesirable LF (??) violates (??) because one occurrence of the variable indexed 2, the antecedent object, is bound by the lambda-binder $\lambda 2$ but another one, the elided object, is not. With (??) in place, a LF that gives rise to the same meaning (??) must contraindex the elided and antecedent objects, but ellipsis will be predicted to be unlicensed for such a LF, a welcome result.

The bound reading of conditional *wh*-questions is a case of a VP Ellipsis configuration known as *Co-binding* (?), where the elided and antecedent objects are bound by the same binder. The solution to the ellipsis licensing problem I described so far for the bound reading has been applied generally to Co-binding examples in the literature.

4.3 The indefinite reading

Accounting for the indefinite reading of conditional *wh*-questions requires some work. Recall that the goal is to explain the second part of the claim (??) from Section ??, i.e., why the indefinite reading is possible iff the *if*-clause is interpreted at a high, question-external position.

At the beginning of Section ??, we envisioned that this explanation would pan out if we could make use of the intuition that *wh*-phrases and indefinites both denote existential quantification. Let us explore this intuition more concretely. Taking the denotation of a question to be a set of propositions that denote possible answers (??), the denotation of a simple *wh*-question is given in (??). On the second line of that equation, the set denotation is rewritten as its characteristic function, which makes it more obvious that existential quantification is involved, just like (??), a minimally different sentence with an indefinite instead of a *wh*-phrase.

$$(43) \quad \begin{aligned} \text{a. } \llbracket \text{who Sally met} \rrbracket^g &= \{\text{met}(x)(\text{sally}) \mid \text{person}(x)\} \\ &= \lambda p_{st}. \exists x_e. \text{person}(x) \wedge p = \text{met}(x)(\text{sally}) \\ \text{b. } \llbracket \text{Sally met someone} \rrbracket^g &= \exists x_e. \text{person}(x) \wedge \text{met}(x)(\text{sally}) \end{aligned}$$

We need to consider how denotations like (??) and (??) are built compositionally. For ease of exposition, I choose to ground the subsequent discussion in an alternative semantics for *wh*-questions and indefinites, but this choice isn't theoretically significant; a similar discussion can be framed within a ? semantics as well.

4.3.1 Alternative semantics

I work with an alternative semantics, whereby the denotations of most linguistic expressions are sets. “Normal” lexical items denote singleton sets, while indefinites and *wh*-phrases denote non-trivial sets of alternatives. I adopt a simplified two-tiered semantics of *wh*-phrases (???), thus distinguishing two kinds of denotations: *ordinary semantic values* ($\llbracket \cdot \rrbracket$) and *focus semantic values* ($\llbracket \cdot \rrbracket_f$). This distinction was implicitly introduced when we adopted (??), the focus condition for ellipsis licensing. In general, lexical items have equal ordinary and focus semantic values, except *wh*-phrases: their ordinary semantic values are undefined (?).

Some lexical entries are illustrated in (??).

$$(44) \quad \begin{aligned} \text{a. } \llbracket \text{Sally} \rrbracket &= \{\text{sally}\}, \llbracket \text{Sally} \rrbracket_f = \{\text{sally}\} \\ \text{b. } \llbracket \text{met} \rrbracket &= \{\text{met}\}, \llbracket \text{met} \rrbracket_f = \{\text{met}\} \\ \text{c. } \llbracket \text{someone} \rrbracket &= \{x \mid \text{person}(x)\}, \llbracket \text{someone} \rrbracket_f = \{x \mid \text{person}(x)\} \\ \text{d. } \llbracket \text{who} \rrbracket &= \text{undefined}, \llbracket \text{who} \rrbracket_f = \{x \mid \text{person}(x)\} \end{aligned}$$

It will be convenient to designate a new type constructor $\{\cdot\}$, with which I will annotate the types of these alternative denotations. If τ is a type, then $\{\tau\}$ is the type of

sets of objects of type τ . Then, $\llbracket \text{Sally} \rrbracket, \llbracket \text{Sally} \rrbracket_f, \llbracket \text{someone} \rrbracket, \llbracket \text{someone} \rrbracket_f$ and $\llbracket \text{who} \rrbracket_f$ all have type $\{e\}$, while $\llbracket \text{met} \rrbracket$ and $\llbracket \text{met} \rrbracket_f$ both have type $\{est\}$ (an intension).

The denotation function is defined for non-lexical trees by the usual syncategorematic rules, except these rules are extended to work with alternatives and to handle semantic computation in both the ordinary and the focus semantic space. For example, Function Application is extended to a rule known as Pointwise Function Application (??).

(45) *Pointwise Function Application*

If A, B are trees, $\llbracket A \rrbracket$ has type $\{\tau_1 \tau_2\}$ for some types τ_1, τ_2 and $\llbracket B \rrbracket$ has type $\{\tau_1\}$, then $\llbracket A B \rrbracket$ has type $\{\tau_2\}$, defined by

$$\llbracket A B \rrbracket^g = \{f(x) \mid f \in \llbracket A \rrbracket^g, x \in \llbracket B \rrbracket^g\}$$

If A, B are trees, $\llbracket A \rrbracket_f$ has type $\{\tau_1 \tau_2\}$ for some types τ_1, τ_2 and $\llbracket B \rrbracket_f$ has type $\{\tau_1\}$, then $\llbracket A B \rrbracket_f$ has type $\{\tau_2\}$, defined by

$$\llbracket A B \rrbracket_f^g = \{f(x) \mid f \in \llbracket A \rrbracket_f^g, x \in \llbracket B \rrbracket_f^g\}$$

Indefinites receive their existential meanings by interacting with $\exists$, an operator associated with the syncategorematic rule (??). Effectively, it flattens a set of propositions by existential quantification into a singleton set containing exactly one proposition. With this rule, we fully derive the denotation of the indefinite sentence (??), as in (??). The rule invokes a function EC, which takes a set of propositions and existentially closes them into a single proposition.

(46) *Existential Closure*

If A is a tree and $\llbracket A \rrbracket$ has type $\{st\}$, then $\llbracket \exists A \rrbracket$ has type $\{st\}$, defined by

$$\llbracket \exists A \rrbracket^g = \{EC(\llbracket A \rrbracket^g)\}$$

If A is a tree and $\llbracket A \rrbracket_f$ has type $\{st\}$, then $\llbracket \exists A \rrbracket_f$ has type $\{st\}$, defined by

$$\llbracket \exists A \rrbracket_f^g = \{EC(\llbracket A \rrbracket_f^g)\}$$

where $EC(P)$ is defined for a set P of propositions as

$$EC(P) = \lambda w_s. \exists p_{st} \in P. p(w)$$

$$\begin{aligned} (47) \quad & \llbracket \exists [\text{Sally} [\text{met someone}]] \rrbracket^g \\ &= \{EC(\llbracket \text{John} [\text{met someone}] \rrbracket^g)\} \\ &= \{\lambda w_s. \exists p_{st} \in \llbracket \text{John} [\text{met someone}] \rrbracket^g. p(w)\} \\ &= \{\lambda w_s. \exists p_{st} \in \{\text{met}(x)(\text{sally}) \mid \text{person}(x)\}. p(w)\} \\ &= \{\lambda w_s. \exists x_e. \text{person}(x) \wedge \text{met}(x)(\text{sally})(w)\} \end{aligned}$$

Wh-phrases lack ordinary semantic values. Observe that the syncategorematic rules for a non-lexical tree, e.g., (??) and (??), require that tree's daughters to have ordinary semantic values in order to calculate an ordinary semantic value for the tree itself. Thus, any constituent that contains a *wh*-phrase would also lack an ordinary semantic value, making *wh*-questions uninterpretable. This is where the question operator Q comes in (??); it transfers the semantic value of its sister from the focus semantic space to the ordinary semantic space. This allows us to interpret a *wh*-question like (??), as in (??).

(48) *Question*

If A is a tree and $\llbracket A \rrbracket_f$ has type $\{st\}$, then $\llbracket Q A \rrbracket, \llbracket Q A \rrbracket_f$ both have type $\{st\}$, defined by

$$\llbracket Q A \rrbracket^g = \llbracket Q A \rrbracket_f^g = \llbracket A \rrbracket_f^g$$

(49) $\llbracket [Q [Sally [met\ who]]] \rrbracket^g$
 $= \llbracket [Sally [met\ who]] \rrbracket_f^g$
 $= \{met(x)(sally) \mid person(x)\}$

Two notes are in order at this point. First, in (??), I assume the *wh*-phrase is interpreted *in situ*. This significantly simplifies derivations, but it isn't necessary. In fact, some examples require *wh*-phrases to be interpreted at moved positions, e.g., in the bound reading of conditional *wh*-questions, where *wh*-movement creates new binding relations. Alternative semantics allow an *ex situ* interpretation of *wh*-phrases, but it requires a slightly more complicated reformulation of the simple system I describe here (?????). Nevertheless, it can be done in a way that doesn't affect our subsequent discussion. In other words, we will not lose our account for the bound reading of conditional *wh*-questions.

The second note is that the operators I use for interpreting *wh*-phrases and indefinites, namely Q and $\exists$, are unselective. Selectivity can be introduced into our system without affecting our subsequent discussion in certain ways, e.g., by indexing alternative generators (???) or by making sets a recursive type (?).

The system I just described derives (??), an important generalization.

(50) A constituent containing a *wh*-phrase that isn't c-commanded by Q will not have an ordinary semantic value.

This generalization is an attractive feature of such a two-tiered semantics for *wh*-questions, as it allows for a technical account of various kinds of intervention effects in *wh*-questions (?????). I will also make crucial use of (??) in my account of the indefinite reading of conditional *wh*-questions.

4.3.2 A partial account of the indefinite reading

Having described the alternative semantics framework I will assume in the rest of this paper, I now turn to the indefinite reading of conditional *wh*-questions. To simplify the discussion, I only consider *wh*-phrases of the form *what NP*. As for the elided indefinite object, it doesn't matter what we assume its exact form to be, which is made undetectable by the ellipsis anyways. For now, I will assume the elided indefinite

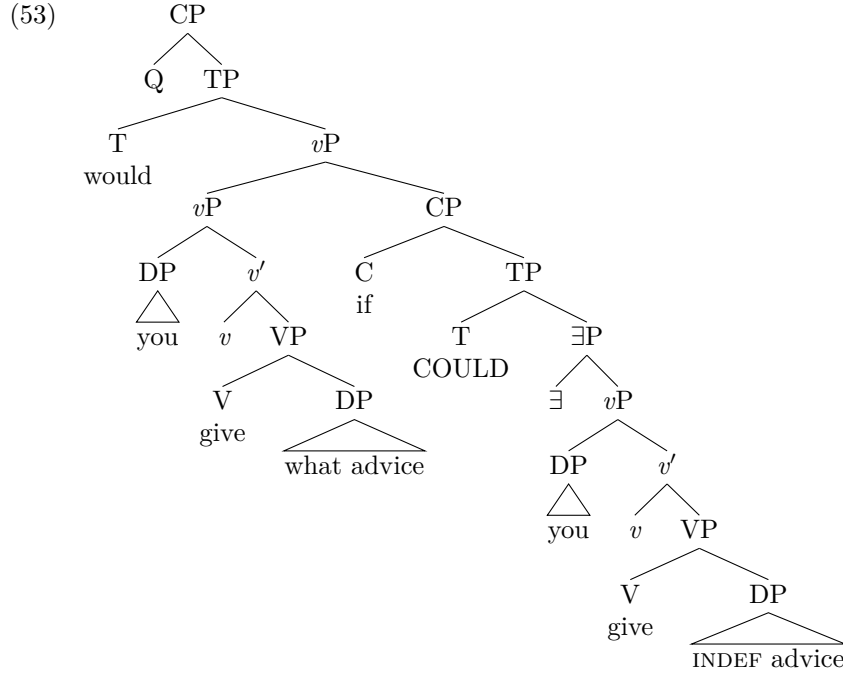
has the form *INDEF NP*, where *INDEF* is a null indefinite determiner. Crucially, an indefinite DP of this form shares a denotation with the focus semantic value of *what NP* (??).

- (51) $\llbracket \text{what NP} \rrbracket^g = \text{undefined}$ $\llbracket \text{what NP} \rrbracket_f^g = \{x \mid \llbracket \text{NP} \rrbracket^g(x)\}$
 $\llbracket \text{INDEF NP} \rrbracket^g = \{x \mid \llbracket \text{NP} \rrbracket^g(x)\}$ $\llbracket \text{INDEF NP} \rrbracket_f^g = \{x \mid \llbracket \text{NP} \rrbracket^g(x)\}$

Since ordinary semantic values are now sets, just like focus semantic values, we need to update the old definition of contrast (??) to (??).

- (52) *E* contrasts with *A* iff the ordinary semantic value $\llbracket A \rrbracket^g$ is a subset of the focus semantic value $\llbracket E \rrbracket_f^g$ for all assignments *g*.

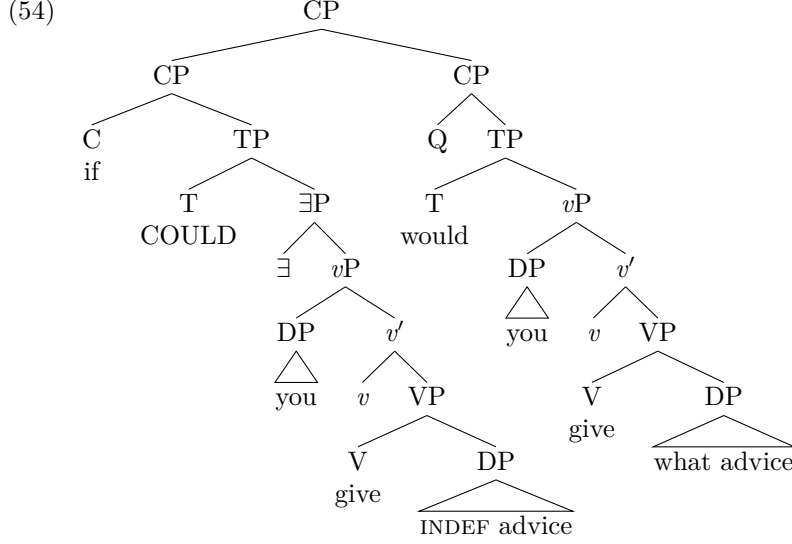
Our adoption of a two-tiered semantics for *wh*-questions gives us an account of why the indefinite reading is impossible with a low *if*-clause. Such a LF is illustrated in (??). This LF is a revision of the LF (??) I presented in Section ?? . The only changes are that (i) the elided indefinite object now contains an indefinite D head *INDEF*, and that (ii) the operator $\exists$ that interacts with the elided indefinite object appears below the modal *COULD*. The position of $\exists$ is crucially responsible for the failure of the present account, and we will discuss it soon.



By (??), ellipsis is licensed only if we can find parallelism domains *A* and *E* such that $\llbracket A \rrbracket^g \subseteq \llbracket E \rrbracket_f^g$ for all assignments *g*. This requires that $\llbracket A \rrbracket^g$ be defined in the first place. Since *A* must contain the antecedent *wh*-object *what advice*. By (??), $\llbracket A \rrbracket^g$ will be undefined unless *A* also contains the question operator *Q* that c-commands the *wh*-object. The only possible *A* is the entire *wh*-question. Then, *A* would contain any

choice of E . This results in an inescapable antecedent containment configuration, as the focus condition can't be satisfied if A properly contains E . Thus, we correctly predict the ellipsis in (??) to be ungrammatical.

Let's turn to an indefinite reading LF with a high *if*-clause (??), which should be grammatical.



We need to locate parallelism domains A and E that satisfy the focus condition. Again, A needs to contain Q in order to have an ordinary semantic value. The *wh*-question headed by Q might be a potential choice for A . An appropriate choice for E might be the TP headed by the modal *COULD*. This choice of A and E are indicated in (??).

- (55) a. $E = [\text{TP}_F [\text{COULD}] \exists \text{ you give INDEF advice}]$
b. $A = [\text{CP } Q \text{ would you give what advice}]$

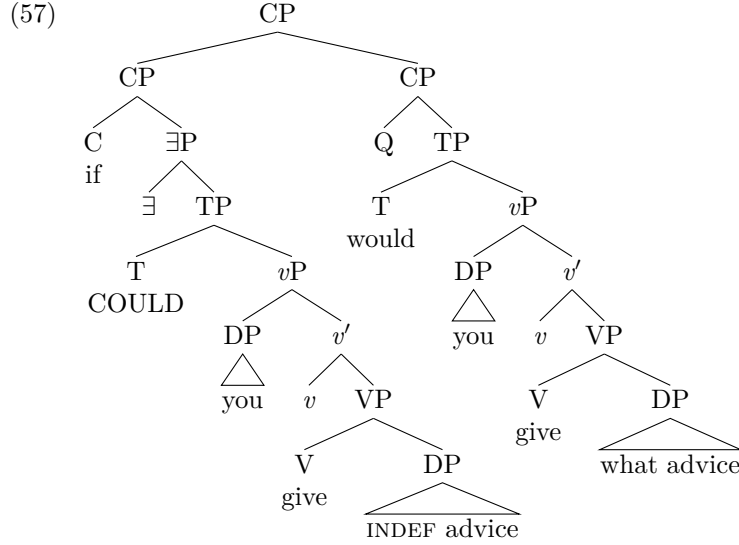
In order for E to contrast with A , all the parts of E responsible for any semantic difference between E and A are focus-marked. The ordinary semantic values of A and E are derived in (??). We see that these are different denotations.

- (56) a. $\llbracket A \rrbracket^g = \llbracket Q \text{ would you give what advice} \rrbracket^g$
 $= \llbracket \text{would you give what advice} \rrbracket_f^g$
 $= \{\text{would}(x) \mid x \in \llbracket \text{you give what advice} \rrbracket_f^g\}$
 $= \{\text{would}(f(\text{you})) \mid f \in \llbracket \text{give what advice} \rrbracket_f^g\}$
 $= \{\text{would}(\text{give}(x)(\text{you})) \mid x \in \llbracket \text{what advice} \rrbracket_f^g\}$
 $= \{\text{would}(\text{give}(x)(\text{you})) \mid \text{advice}(x)\}$
b. $\llbracket E \rrbracket^g = \llbracket \text{F}[\text{COULD}] \exists \text{ you give INDEF advice} \rrbracket^g$
 $= \{\text{could}(x) \mid x \in \llbracket \exists \text{ you give INDEF advice} \rrbracket^g\}$
 $= \{\text{could}(\text{EC}(\llbracket \text{you give INDEF advice} \rrbracket^g))\}$
 $= \{\text{could}(\text{EC}(\{\text{give}(x)(\text{you}) \mid \text{advice}(x)\}))\}$

There are two ways in which *E* and *A* semantically differ. First, there is a lexical difference between the modals: *E* has *COULD*, while *A* has *would*. Second, the operator $\exists$ is present in *E* but not in *A*. As a result, the alternatives propagated from the indefinite *INDEF advice* are existentially closed in *E* by EC (??), but the alternatives propagated from the *wh*-phrase *what advice* are left unclosed in *A* (??).

The lexical difference between *COULD* in *E* and *would* in *A* can be neutralized by focus-marking *COULD*. This correctly accounts for the obligatory accent on *COULD*; it is the realization of a focus-mark that is necessary for licensing the ellipsis in (??).

However, the semantic difference caused by $\exists$ prevents *E* and *A* from satisfying the focus condition. This results in the incorrect prediction that the ellipsis in (??) is ungrammatical. This problem could be circumvented if $\exists$ were placed above *COULD* in the *if*-clause, i.e., if the elided indefinite had wide scope with respect to the modal. That LF would be (??).



In this case, we would end up with the selection of *E* and *A* as in (??).

- (58) a. *E* = [_{TP} F[*COULD*] you give *INDEF advice*]
b. *A* = [_{CP} Q *would* you give what advice]

Then, the only semantic difference between *E* and *A* would be the lexical difference between *COULD* and *would*, which is neutralized by focus-marking. The ellipsis in (??) would be correctly predicted to be licensed.

However, we know that the elided indefinite in (??) takes narrow scope from examples like (??), adapted from the examples in Section ?? . In this example, the modal in the *if*-clause is a necessity modal, which allows us to tease apart the interpretations generated from the narrow- vs. wide-scope placement of $\exists$; narrow scope gives rise to a non-specific reading of the indefinite, while wide scope gives rise to a specific reading. As a matter of fact, the elided indefinite only allows a non-specific (??) and not a specific (??) reading. It thus appears that $\exists$ must take narrow scope, i.e., be placed under *COULD* in (??).

- (59) If Bill were asked to Δ , what Kraftwerk song would he play?
- (60) a. *Narrow scope (non-specific)*
 If Bill were asked $\exists$ to $\langle \text{play-IND} \text{ Kraftwerk song} \rangle$, what Kraftwerk song would he play?
 Meaning: ‘If Bill were asked to play a Kraftwerk song of his choice, what Kraftwerk song would he play?’
- b. *Wide scope (specific)*
 *If $\exists$ Bill were asked to $\langle \text{play-IND} \text{ Kraftwerk song} \rangle$, what Kraftwerk song would he play?
 Meaning: ‘If there is a specific Kraftwerk song that Bill were asked to play, what Kraftwerk song would he play?’

Let’s summarize the problem at hand. In order to correctly account for the obligatory accent on *COULD*, it seems desirable to require $\exists$ to take wide scope with respect to the modal *COULD* in (??). However, it seems necessary to assume $\exists$ has narrow scope, if we want to capture the non-specific reading of the indefinite. We thus have conflicting analytical tensions to model the indefinite as a wide-scope or a narrow-scope existential. Next, I show that this conflict is resolvable with an analysis of *wh*-phrases and indefinites based on Skolemized choice functions.

4.3.3 Skolemized choice functions

A Skolemized choice function is a choice function with an extra argument that parametrizes its choices. For any type τ , a τ -Skolemized choice function is a function f of type $\langle et, \tau e \rangle$ that satisfies $X(f(X)(y))$ for any non-empty set of entities $X \in D_{et}$ and an object y of type τ . I write SCF_τ for the set of τ -Skolemized choice functions.

Skolemized choice functions allow us to describe “narrow-scope” behaviors of indefinites and *wh*-phrases using wide-scope existential quantification. For example, (??) has a reading which is verified by a context where the students each read a different paper. This reading can be described by the LF (??), where the indefinite *some paper* denotes an entity variable y that is bound by a narrow-scope existential quantifier. But it can also be described by the LF (??), where the same indefinite denotes an e -Skolemized choice function variable f that is existentially quantified with wide scope. e is the type of entities. An f that satisfies the existential quantification might pick out different papers for different students: $f(\text{paper})(\text{john})$ could be John’s paper, $f(\text{paper})(\text{sally})$ could be Sally’s paper, and so on.

- (61) Every student read some paper.
- (62) a. $\forall x_e. \text{student}(x) \rightarrow \exists y_e. \text{paper}(y) \wedge \text{read}(y)(x)$
 b. $\exists f_{\langle et, ee \rangle}. f \in SCF_e \wedge \forall x_e. \text{student}(x) \rightarrow \text{read}(f(\text{paper})(x))(x)$

A Skolemized choice functional analysis of indefinites and *wh*-phrases are not only possible, but necessary, as independent facts show. For example, *wh*-questions are known to admit functional answers (??), as shown in (??). For convenience, we may assume the fragment answer in (??) is an elided form of the proposition (??).

- (63) a. Q: Which integer does every integer precede?
 A: Its successor.
 b. Every integer precedes its successor.

The answer in (??) is not referential: a successor isn't a specific integer. Rather, a successor is a function that picks, for every integer n , some integer that is preceded by n . If the denotation of a *wh*-question is a set of possible (propositional) answers, it has to include the unelided proposition (??) as one member. This requirement isn't met in (??), which is the denotation we would produce for the *wh*-question if *wh*-phrases range over individuals x . However, we can write a denotation like (??), which assumes that *wh*-phrases range over Skolemized choice functions f . This set contains (??), because a successor is a Skolemized choice function: consider the function *succ* defined by $succ(integer)(n) = n + 1$, where $n \in integer$.

- (64) a. $\{\forall y_e. integer(y) \rightarrow precedes(x)(y) \mid integer(x)\}$
 b. $\{\forall y_e. integer(y) \rightarrow precedes(f(integer)(y))(y) \mid f \in SCF_e\}$

As for indefinites, consider ?'s discussion of indefinites headed by the phrase *a certain*. Generally, *a certain* behaves as a wide-scope existential. For example, (??) shows that indefinites headed by *a certain* can introduce referents that antecede a pronoun in a future sentence.

- (65) Richard does not have time to date a certain woman, but he sends her flowers.

However, Hintikka shows that *a certain* shows narrow-scope behavior in certain cases. In (??), the referents of *a certain NP* seems to covary with a quantifier in each sentence. How can the apparently contradictory scopal behaviors of *a certain* be reconciled?

- (66) a. According to Freud, every man unconsciously wants to marry a certain woman – his mother.
 b. Each husband had forgotten a certain date – his wife's birthday.

? analyzes *a certain* as denoting Skolemized choice functions. Recall that a Skolemized choice function takes two arguments: a set of entities to choose from, and an extra argument to parametrize the choice with. Kratzer assumes that the extra argument is an implicit bound variable, typical of so-called “anaphoric” or “dependent” predicates like *local*, *enemy*, *foreigner*, etc. (??). The Skolemized choice function itself is existentially quantified with widest scope, but the parameter argument of *a certain* may be bound by other quantifiers, e.g., *every man* and *each husband* in (??), creating an apparent narrow-scope reading. The resulting denotations for (??) are given in (??).

- (67) a. $\exists f_{\langle et, ee \rangle} \in SCF_e. \forall x_e. man(x) \rightarrow want(marry(f(woman)(x))(x))(x)$
 b. $\exists f_{\langle et, ee \rangle} \in SCF_e. \forall x_e. husband(x) \rightarrow had-forgotten(f(date)(x))(x)$

The analytical tensions we face for the elided indefinite in conditional *wh*-questions are mirrored in our discussion of *a certain*. We want *a certain* to take widest scope due to its general wide-scope properties. However, we also want it to scope below a quantifier because its interpretation can covary with that of the quantifier, even though

this covariance is not realized with an overt bound pronoun. Skolemized choice functions allow us to sever the restrictor of an indefinite from its existential quantifier, and effectively parametrize the restrictor with an extra variable. The existential quantification can be wide-scope, while the restrictor sits low enough for its extra variable to be bound by the right quantifier.

So far, we've only looked at *e*-Skolemized choice functions, *e* being the type of entities. The extra argument in Skolemized choice function can be of other types. For example, writing *s* for the type of possible worlds, *s*-Skolemized choice functions can be used to model indefinites that take narrow scope with respect to modals and intensional predicates. An example denotation (??) is given for (??). Here, *f* might pick out a different song in each possible world accessible via the modal *must*.

(68) Bill must play a song.

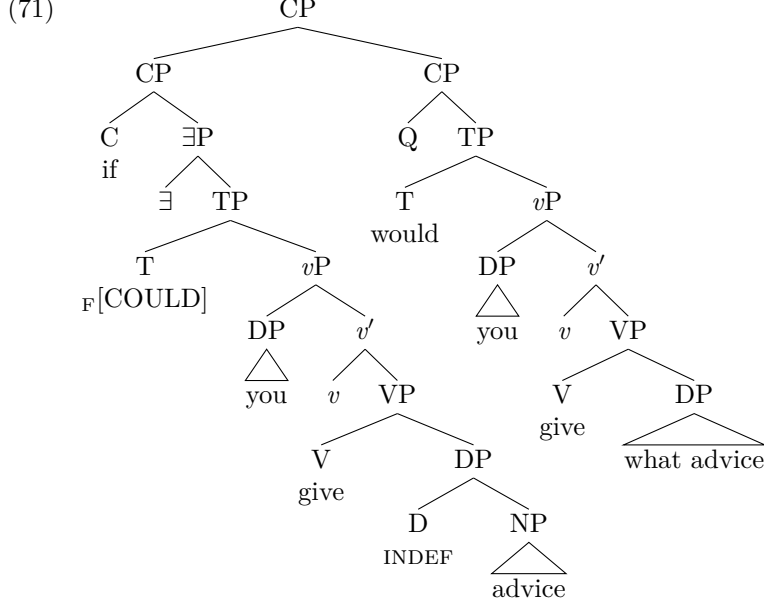
(69) $\exists f_{\langle et, se \rangle}. f \in SCF_s \wedge \text{must}(\lambda w_s. \text{play}(f(\text{song})(w))(\text{bill})(ws))$

Choice functions are 1-ary functions, while τ -Skolemized choice functions are 2-ary functions; τ determines the type of the extra argument. Given types $\tau_1, \dots, \tau_n$, we can generalize Skolemized choice functions to $(n + 1)$ -ary functions for any $n \geq 1$.

We are ready to apply Skolemized choice functions to the elided indefinite object in the indefinite readings of conditional *wh*-questions. Exactly which class of Skolemized choice functions we will use depends on the number of scope-taking operators the indefinite scopes under, and their types. In the familiar example (??), the elided indefinite object takes narrow scope with respect to one scope-taking operator, namely the modal *COULD*. Thus, we will use *s*-Skolemized choice functions for (??). I assume that the *wh*-determiner *what* (that combines with a NP) and the indefinite determiner *INDEF* denotes the appropriate class of Skolemized choice functions (??). Following the two-tiered analysis of *wh*-phrases, we leave $\llbracket \text{what} \rrbracket^g$ undefined. These denotations allow *what* and *INDEF* to combine with their NP complements via Pointwise Function Application.

(70) $\llbracket \text{what} \rrbracket^g = \text{undefined} \quad \llbracket \text{what} \rrbracket_f^g = SCF_s$
 $\llbracket \text{INDEF} \rrbracket^g = SCF_s \quad \llbracket \text{INDEF} \rrbracket_f^g = SCF_s$

I propose a new LF for the indefinite reading with a high *if*-clause, (??). This differs from the earlier LF (??) crucially in the position of $\exists$; here, it is above the modal *COULD*.



The denotations of the *wh*-question and the *if*-clause are given in (??).

$$\begin{aligned}
 (72) \quad & \llbracket \text{Q would you give what advice} \rrbracket^g \\
 &= \{ \text{would}(\lambda w_s. \text{give}(w)(f(\text{advice})(w))(you)) \mid f \in SCF_s \} \\
 & \quad \llbracket \text{if } \exists_F[\text{COULD}] \text{ you give INDEF advice} \rrbracket^g \\
 &= \{ \text{if}(\text{EC}(\{ \text{could}(\lambda w_s. \text{give}(w)(f(\text{advice})(w))(you)) \mid f \in SCF_s \})) \} \\
 &= \{ \text{if}(\lambda w_s. \exists f_{\langle et, se \rangle} \in SCF_s. \text{could}(w)(\lambda w'_s. \text{give}(w')(f(\text{advice})(w'))(you))) \}
 \end{aligned}$$

Finally, we need to look for a pair of parallelism domains to license ellipsis for this LF. We may set A to be the entire *wh*-question headed by Q , and E to be the TP headed by $COULD$. Since A contains Q in addition to a *wh*-phrase, it has an ordinary semantic value. I show $\llbracket A \rrbracket^g$ and $\llbracket E \rrbracket_f^g$ in (??). The set $D_{\langle st, st \rangle}$ represents the set of propositional modifiers, which crucially includes *would* and *could*.

$$\begin{aligned}
 (73) \quad & \llbracket A \rrbracket^g = \llbracket \text{Q would you give what advice} \rrbracket^g \\
 &= \{ \text{would}(\lambda w_s. \text{give}(w)(f(\text{advice})(w))(you)) \mid f \in SCF_s \} \\
 & \quad \llbracket E \rrbracket_f^g = \llbracket \text{if } \exists_F[\text{COULD}] \text{ you give INDEF advice} \rrbracket^g \\
 &= \{ h(\lambda w_s. \text{give}(w)(f(\text{advice})(w))(you)) \mid f \in SCF_s, h \in D_{\langle st, st \rangle} \} \\
 &= \{ \text{would}(\lambda w_s. \text{give}(w)(f(\text{advice})(w))(you)) \mid f \in SCF_s \} \cup \\
 & \quad \{ \text{could}(\lambda w_s. \text{give}(w)(f(\text{advice})(w))(you)) \mid f \in SCF_s \} \cup \\
 & \quad \dots
 \end{aligned}$$

Thus, $\llbracket A \rrbracket_g$ is a subset of $\llbracket E \rrbracket_f^g$, which means the focus condition holds. Ellipsis is correctly predicted to be licensed. This concludes my account for the indefinite reading of conditional *wh*-questions.

5 VP Ellipsis in other kinds of adjuncts

In this section, I explore a prediction made by my discussion of the indefinite reading of conditional *wh*-questions in this paper. What I have been calling “the indefinite reading” is a pattern in VP Ellipsis that allows a *wh*-phrase to antecede an elided indefinite, and in this paper I study the occurrence of the indefinite reading in conditional *wh*-questions, a kind of construction where the elided VP is hosted in a specific environment: *if*-clauses. Using structural diagnostics, I have shown in Section ?? that the attachment height of the *if*-clause controls the availability of the indefinite reading, and attempted to provide an analysis that derives this fact in Section ?. However, nothing about my diagnostics or my analysis says that the elided VP has to be hosted specifically in an *if*-clause, of all kinds of clausal adjuncts, in order for the indefinite reading to surface. Thus, we would expect to see the indefinite reading, i.e., a *wh*-phrase anteceding an elided indefinite, in clausal adjuncts other than *if*-clauses.

Before exploring this prediction, I want to restate it by connecting it to Liliane Haegeman’s work on the external syntax of clausal adjuncts. In a series of papers, ?????? argues that adverbial clauses can be classified into two groups, *central* and *peripheral*, which differ in both their semantic contributions to their matrix clauses as well as the heights at which they are attached in the structure. It is possible for the same subordinating conjunction to introduce a central as well as a peripheral adverbial clause. For example, the *if*-clause in (??) is a central adverbial clause and that in (??) is a peripheral adverbial clause, per Haegeman’s classification.

- (74) a. If it rains, I’ll bring an umbrella.
 b. If it’s going to rain, why don’t we stay at home?

According to Haegeman, central adverbial clauses modify the event structure described by their matrix clauses, while peripheral adverbial clauses modify the discourse structure their matrix clauses are situated in. For example, the central *if*-clause in (??) describes a condition that restricts the umbrella-bringing event described by the matrix clause, while the peripheral *if*-clause in (??) provides a premise that makes the speech act of asking the matrix question relevant and felicitous. This semantic difference between central and peripheral adverbial clauses correlate with their attachment heights relative to their matrix clauses. Haegeman uses structural diagnostics to show that central adverbial clauses are interpreted as attached inside their matrix clauses, while peripheral ones aren’t. To state my results about conditional *wh*-questions in Section ?? in Haegeman’s terms, a conditional *wh*-question has a bound reading when it has a low, central *if*-clause, and an indefinite reading when it has a high, peripheral *if*-clause.

As I said earlier, the availability of the indefinite reading is controlled only by the height of the adjunct that contains the elided VP; the type of the adjunct does not matter. Then, we might expect that the generalization concerning the distribution of bound and indefinite readings to carry over to other kinds of adverbial clauses. For example, Haegeman shows that *while* can introduce central and peripheral adverbial clauses. In (??), *while* is central; it introduces a temporal adjunct. In (??), *while* is peripheral; it introduces a relevant fact that contrasts with the proposition expressed in the matrix clause.

- (75) a. These kids only behave while their parents are watching them.
 b. While I know Bill isn't going to bring anything to the cookout, I wonder what Sally is planning to bring to the cookout.

? shows that central *while*-clauses are structurally integrated in their matrix clauses, while peripheral *while*-clauses aren't. We would then expect central *while*-clauses to be able to host a bound reading and peripheral *while*-clauses to be able to host an indefinite reading of an elided VP. This prediction is borne out in the example (??) below.

- (76) a. (*Context: John, Sally and Bill sat in the same row during a long flight. Sally watched a bunch of movies during the flight, and Bill knows John was looking at her screen from time to time. Bill wants to know which movies Sally watched while John was looking at her screen, and he asks the question in a funny way:*)
 [Which movie]¹ did Sally watch t_1 while John did ~~<watch it>~~_T?
 b. *Context: I reminded you that we are going to our friend Bill's housewarming party. I ask you:*
 While Bill made it clear that we don't have to ~~<bring a gift>~~, [what kind of gift]¹ do you think we should bring t_1 ?

Two notes are in order before I conclude this section. First, I don't think Haegeman's distinction between central and peripheral adverbial adjuncts is the only factor that determines the availability of a bound or an indefinite reading of the elided object anteceded by a *wh*-phrase. However, I do think this distinction makes a good starting point for a future investigation of this kind of VP Ellipsis in other sorts of adverbial clauses.

Second, I have characterized the *if*-clauses of conditional *wh*-questions with the indefinite reading as being "high" and "question-external", but a more accurate characterization might have "not low" and "not question-internal". After all, all that matters is that such *if*-clauses are not attached question-internally; they might not be attached to the *wh*-question at all, and instead might constitute a separate structure from the *wh*-question. Haegeman makes a similar point: while central adverbial clauses are attached inside their matrix clauses, peripheral adverbial clauses aren't necessarily attached outside; all facts merely suggest that they are not integrated into the structure of their matrix clauses. This is an important clarification because I am agnostic to the exact syntax/semantic analysis of peripheral *if*-clauses that result in an indefinite reading of conditional *wh*-questions. Such analyses might differ in whether they take these *if*-clauses to be attached to the *wh*-question or not attached at all; an analysis of either kind should be compatible with the results in this paper.

6 Conclusion

In this paper, I describe a construction I call conditional *wh*-questions with VP Ellipsis, which are root object *wh*-questions that appear with an *if*-clause that hosts an elided VP. The object of the elided VP has two possible interpretations: the bound reading,

whereby the elided object is a variable bound by the *wh*-phrase, or the indefinite reading, whereby the elided object is a narrow-scope indefinite.

Structural diagnostics indicate that the interpretation of the elided object is conditioned by the height at which the *if*-clause is attached: the bound reading is possible only when the *if*-clause is attached question-internally, and the indefinite reading is possible only when the *if*-clause is attached question-externally.

An account of the VP Ellipsis in conditional *wh*-questions must explain two things: why the indefinite reading is possible at all, even though it features a syntactic mismatch between the objects of the antecedent and elided VPs; and why the two interpretations of the elided object depend on different heights of the *if*-clause. Theories of VP Ellipsis are typically divided into those that employ a syntactic identity condition and those that employ a semantic one. I take the apparent syntactic mismatch inherent to the indefinite reading to support a semantic identity condition rather than a syntactic one. However, the connection between the two readings and *if*-clause height prove to be difficult to account for with a semantic identity condition. This is because the indefinite reading of conditional *wh*-questions features a scopal mismatch between the elided object and its antecedent: the antecedent object is a *wh*-phrase, which is an indefinite that takes maximal scope, while the elided object is a narrow-scope indefinite. Analyzing *wh*-phrases and indefinites as denoting Skolemized choice functions allows us to avoid this scope problem.

I end by connecting my discussion of conditional *wh*-questions to Haegeman's work on the distinction between the so-called central and peripheral adverbial clauses. This connection leads me to predict that the bound and indefinite readings are not unique to *if*-clauses and extend to other adverbial clauses; specifically, central adverbial clauses should allow a bound reading, while peripheral adverbial clauses should allow an indefinite reading. I give one example where this prediction seems to be borne out, but a more systematic investigation is needed for me to draw any conclusions.

References

- Abels K (2022) On vehicle change and ellipsis identity. URL <https://lingbuzz.net/lingbuzz/006709>, 006709
- Aelbrecht L (2010) The syntactic licensing of ellipsis. John Benjamins
- Beck S (1996) Quantified structures as barriers for *lf* movement. *Natural Language Semantics* 4:1–56
- Beck S (2006) Intervention effects follow from focus interpretation. *Natural Language Semantics* 14:1–56
- Beck S, Kim SS (1997) On *wh*- and operator scope in Korean. *Journal of East Asian Linguistics* 6:339–384
- Bhatt R, Pancheva R (2006) Conditionals. In: Everaert M, van Riemsdijk H (eds) *The Blackwell Companion to Syntax*. Wiley Online Library, p 638–687

- Bumford D (2022) Composition under distributive natural transformations: Or, when predicate abstraction is impossible. *Journal of Logic, Language, and Information* 31(3):287–307
- Bumford D, Charlow S (to appear) Effect-driven interpretation: Functors for natural language composition. Accepted with minor revisions, Cambridge University Press *Elements in Semantics*, URL <https://ling.auf.net/lingbuzz/008912>
- Charlow S (2020) The scope of alternatives: indefiniteness and islands. *Linguistics and Philosophy* 43(4):427–472
- Charlow S (2021) $\exists$ -closure and alternatives. *Linguistic Inquiry* 52(1):143–152
- Chung S (2013) Syntactic identity in sluicing: How much, and why. *Linguistic Inquiry* 44:1–39
- Chung S, Ladusaw WA, McCloskey J (1995) Sluicing and logical form. *Natural Language Semantics* 3:239–282
- Collins C (1998) A note on extraction from conditionals. *Cornell Working Papers in Linguistics* 16:57–66
- van Craenenbroeck J (2010) The syntax of ellipsis: Evidence from Dutch dialects. Oxford University Press
- Engdahl E (1980) The syntax and semantics of questions in swedish. PhD thesis
- Engdahl E (1986) Constituent questions: The syntax and semantics of questions with special reference to Swedish. Springer
- Fiengo R, May R (1994) *Indices and Identity*. MIT Press
- Fox D (2000) *Economy and semantic interpretation*. MIT Press
- Haegeman L (1984a) Parasitic gaps and adverbial clauses. *Journal of Linguistics* 20(2):229–232. <https://doi.org/10.1017/S0022226700013864>
- Haegeman L (1984b) Pragmatic conditionals in english. *Folia Linguistica* 18(3-4):485–502. <https://doi.org/doi:10.1515/flin.1984.18.3-4.485>
- Haegeman L (1984c) Remarks on adverbial clauses and definite np-anaphora. *Linguistic Inquiry* 15(4):712–715
- Haegeman L (2002) Anchoring to speaker, adverbial clauses and the structure of cp. *Georgetown University Working Papers in Theoretical Linguistics* 2(1):117–180
- Haegeman L (2003a) Conditional clauses: External and internal syntax. *Mind & Language* 18(4):317–339. <https://doi.org/https://doi.org/10.1111/1468-0017.00230>

- Haegeman L (2003b) Speculations on adverbial fronting and the left periphery. In: Guéron J, Tasmowski L (eds) *Temps et point de vue/Tense and point of view*. p 329–365
- Haegeman L (2006) Conditionals, factives and the left periphery. *Lingua* 116(10):1651–1669. <https://doi.org/https://doi.org/10.1016/j.lingua.2005.03.014>
- Hagstrom P (1998) Decomposing questions. PhD thesis
- Hamblin CL (1973) Questions in montague english. *Foundations of Language* 10(1):41–53
- Haspelmath M (1997) *Indefinite Pronouns*. Oxford University Press
- Heim I (1997) Predicates or formulas? evidence from ellipsis. In: Lawson A (ed) *Proceedings of the 7th Semantics and Linguistic Theory Conference (SALT)*, pp 197–221
- Hintikka J (1986) The semantics of *a certain*. *Linguistic Inquiry* 17(2):331–336
- Hoji H (1985) Logical form constraints and configurational structures in japanese. PhD thesis
- Hoji H (1986) Scope interpretation in japanese and its theoretical implications. In: Dalrymple M, Goldberg J, Hanson K, et al (eds) *Proceedings of the 5th West Coast Conference on Formal Linguistics (WCCFL)*. CSLI Publications, pp 87–101
- Iatridou S (1991) Topics in conditionals. PhD thesis, MIT
- Izvorski R (1996) The syntax and semantics of correlative proforms. In: Kusumoto K (ed) *Proceedings of North East Linguistic Society* 26, pp 133–147
- Johnson K (1991) Object positions. *Natural Language & Linguistic Theory* 9(4):577–636
- Johnson K (2001) *What VP Ellipsis Can Do, and What it Can't, But Not Why*, John Wiley & Sons, Ltd, chap 14, pp 439–479
- Karttunen L (1977) Syntax and semantics of questions. *Linguistics and Philosophy* 1(1):3–44
- Kehler A (2002) *Coherence in discourse*. CSLI Publications
- Kennedy C (1997) Vp-deletion and “nonparasitic” gaps. *Linguistic Inquiry* 28(4):697–707
- Kim S, Lyle J (1996) Parasitic gaps, multiple questions, and vp ellipsis. In: Camacho J, Choueiri L, Watanabe M (eds) *Proceedings of the 14th West Coast Conference*

- on Formal Linguistics (WCCFL). CSLI Publications, pp 287–301
- Koizumi M (1993) Object agreement phrases and the split vp hypothesis. In: MIT Working Papers in Linguistics, vol 18. p 99–148
- Kotek H (2019) On the semantics of *wh*-questions. In: Proceedings of Sinn Und Bedeutung, 20, pp 430–447
- Kratzer A (1991) The Representation of Focus, De Gruyter Mouton, pp 825–834
- Kratzer A (1998) Scope or pseudoscope? are there wide-scope indefinites? In: Rothstein S (ed) Events and Grammar: Studies in Linguistics and Philosophy 70. Springer
- Kratzer A, Shimoyama J (2002) Indeterminate pronouns: The view from japanese. In: Otsu Y (ed) The Proceedings of the Third Tokyo Conference on Psycholinguistics. Hituzi Syoboo, pp 1–25
- Merchant J (2001) The syntax of silence: Sluicing, islands, and the theory of ellipsis. Oxford University Press
- Merchant J (2013) Voice and ellipsis. *Linguistic Inquiry* 44:77–108
- Merchant J (2018) Verb-stranding predicate ellipsis in greek, implicit arguments, and ellipsis-internal focus. In: A Reasonable Way to Proceed: Essays in Honor of Jim McCloskey. UCSC Linguistics Research Center
- Mitchell JE (1987) The formal semantic point of view. PhD thesis
- Partee B (1989) Binding implicit variables in quantified contexts. In: Wiltshire C, Graczyk R, Music B (eds) Papers from the 25th Annual Regional Meeting of the Chicago Linguistic Society, pp 1–24
- Reinhart T (1983) Coreference and bound anaphora: A restatement of the anaphora questions. *Linguistics and Philosophy* 6(1):47–88
- van Riemsdijk H, Williams E (1981) Np-structure. *The Linguistic Review* 1(2):171–218
- Romero M, Novel M (2013) Variable binding and sets of alternatives, Palgrave Macmillan UK, pp 174–208
- Sag IA, Hankamer J (1984) Toward a theory of anaphoric processing. *Linguistics and Philosophy* 7:325–345
- Schwarzschild R (1999) Givenness, avoidf and other constraints on the placement of accent. *Natural Language Semantics* 7:141–177

- Shan Cc (2004) Binding alongside hamblin alternatives calls for variable-free semantics. In: Young RB (ed) Proceedings of the 14th Semantics and Linguistic Theory Conference (SALT), pp 289–304
- Takahashi S, Fox D (2005) Maxelide and the re-binding problem. In: Georgala E, Howell J (eds) Proceedings of the 15th Semantics and Linguistic Theory Conference (SALT), pp 223–240
- Uegaki W (2018) A unified semantics for the japanese q-particle *ka* in indefinites, questions and disjunctions. *Glossa: a journal of general linguistics* 3(1)
- Williams ES (1977) Discourse and logical form. *Linguistic Inquiry* 8(1):101–139
- Wold DE (1996) Long distance selective binding: The case of focus. In: Galloway T, Spence J (eds) Proceedings of the 6th Semantics and Linguistic Theory Conference (SALT), pp 311–328